Assignment 7

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PROBLEM STATEMENT

Develop a web-based application such as an e-commerce platform, student feedback system, or similar solution using MongoDB as the backend, along with React.js and Express.js.

AIM

To design and implement a web application called EduPulse – Course Feedback and Review System using the MERN stack (MongoDB, Express.js, React.js, Node.js). This system allows students to submit structured feedback on courses, helping educational institutions assess and enhance teaching quality.

OBJECTIVES

1. To create an interactive and user-friendly interface using React.js
2. To implement secure user authentication using JWT
3. To develop a robust and scalable backend system
4. To efficiently store and manage data using MongoDB
5. To enable analysis of feedback for continuous improvement

THEORY

1. Introduction

EduPulse is a web-based feedback system built using the MERN stack. It enables students to provide structured feedback on courses, allowing institutions to evaluate and improve teaching standards.
Traditional feedback methods are often manual and inefficient, lacking real-time insights. EduPulse addresses these issues by providing a digital, scalable, and easy-to-use platform for collecting and analyzing feedback.

2. Technology Stack

Frontend – React.js

- Used for building a dynamic and responsive user interface
- Based on a component-driven architecture
- Manages user interaction and data display

- Utilizes hooks such as useState and useEffect

Backend – Node.js & Express.js

- Node.js serves as the runtime environment
- Express.js is used to handle routing and API development
- REST APIs enable communication between frontend and backend
- Responsible for authentication, validation, and core logic

Database – MongoDB Atlas

- A NoSQL database storing data in JSON-like format
- Offers flexible schema design
- Stores data such as:
    - Users
    - Courses
    - Feedback

3. System Architecture

Workflow:

1. User interacts with the React frontend
2. Frontend sends API requests to the backend
3. Backend processes requests using Express.js
4. Data is stored or retrieved from MongoDB
5. The response is returned to the frontend

4. System Modules

1. User Authentication Module

- Handles registration and login
- May include password hashing
- Uses JWT for authentication
- Ensures secure access to the system

2. Course Management Module

- Allows adding and viewing courses
- Stores course information in the database
- Displays available courses to users

3. Feedback Module

- Enables students to submit feedback
- Feedback is linked to specific courses
- Includes:
    - Ratings
    - Comments

4. Data Storage Module

- Uses MongoDB collections:
    - users
    - courses
    - feedback

5. API Module

- Provides RESTful endpoints such as:
    - /api/auth
    - /api/courses
    - /api/feedback

5. System Working (Step-by-Step)

1. User registers or logs in
2. A JWT token is generated
3. User accesses the list of courses
4. Feedback is submitted
5. Backend validates the request
6. Data is stored in MongoDB
7. Feedback is retrieved and displayed

6. JWT Authentication

- JWT (JSON Web Token) is used for secure communication

Process:

1. User logs in
2. Server generates a token using a secret key
3. Token is sent to the client
4. Client includes the token in future requests
5. Server verifies the token

This ensures that only authorized users can access the system.

7. Advantages

- Enables real-time feedback collection
- Scalable due to MongoDB
- Easy-to-use interface
- Secure authentication mechanism
- Efficient data retrieval and analysis

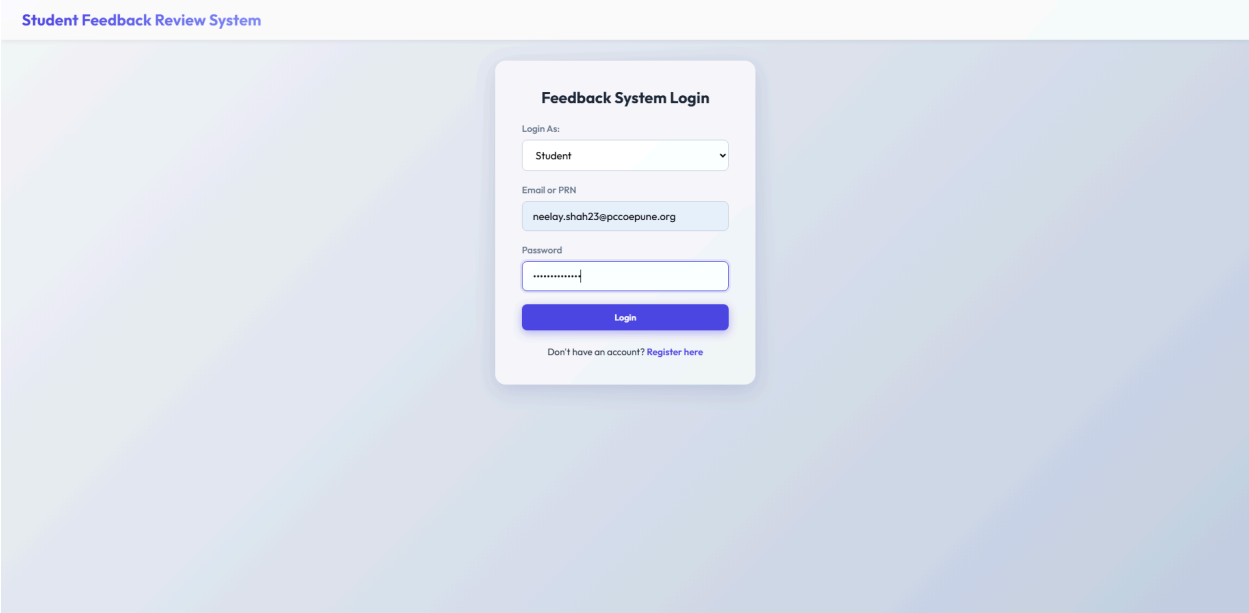
8. Limitations

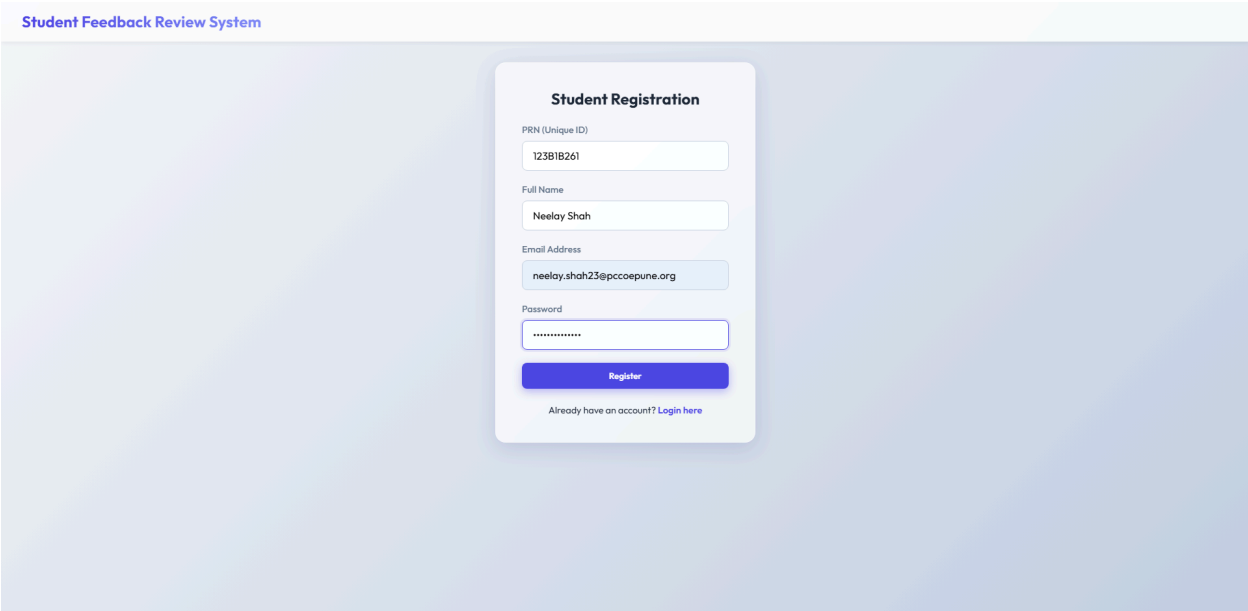
- Requires an internet connection
- Provides only basic analysis (no advanced AI features)
- Relies on honest user input

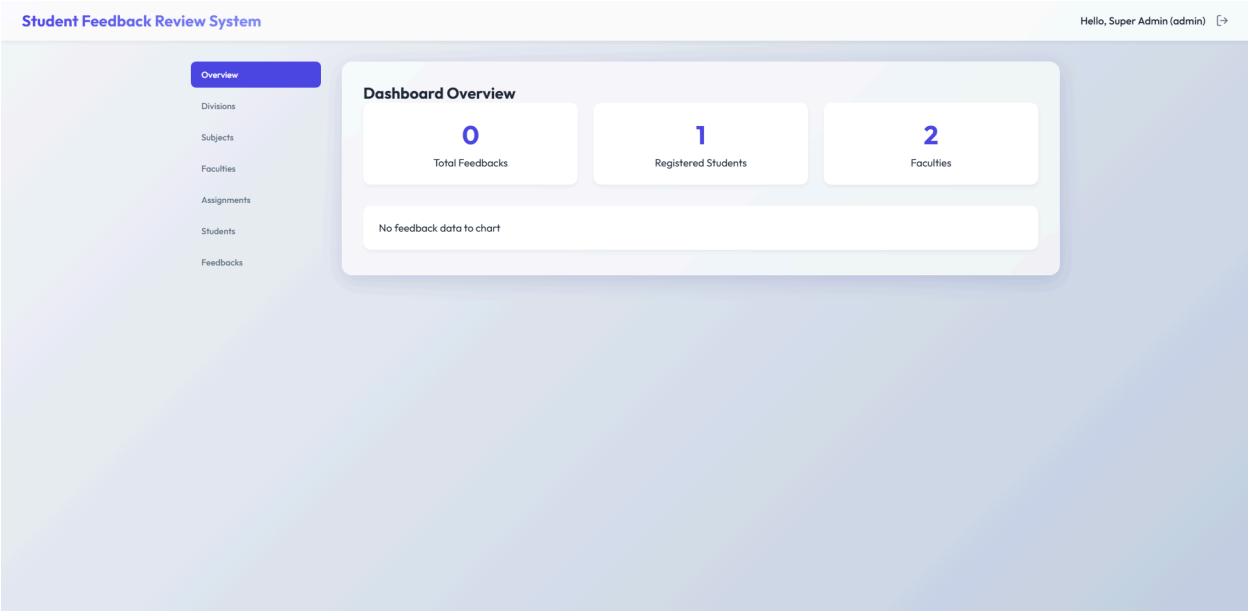
9. Applications

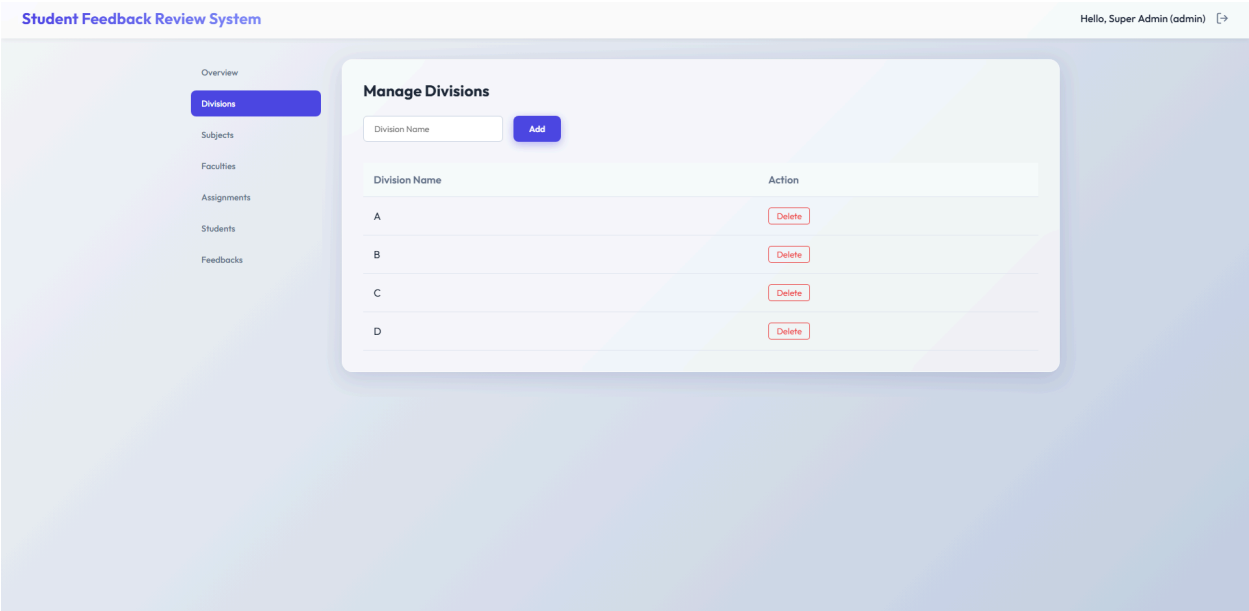
- Educational institutions (colleges and universities)
- Online learning platforms
- Training centers
- Corporate training programs

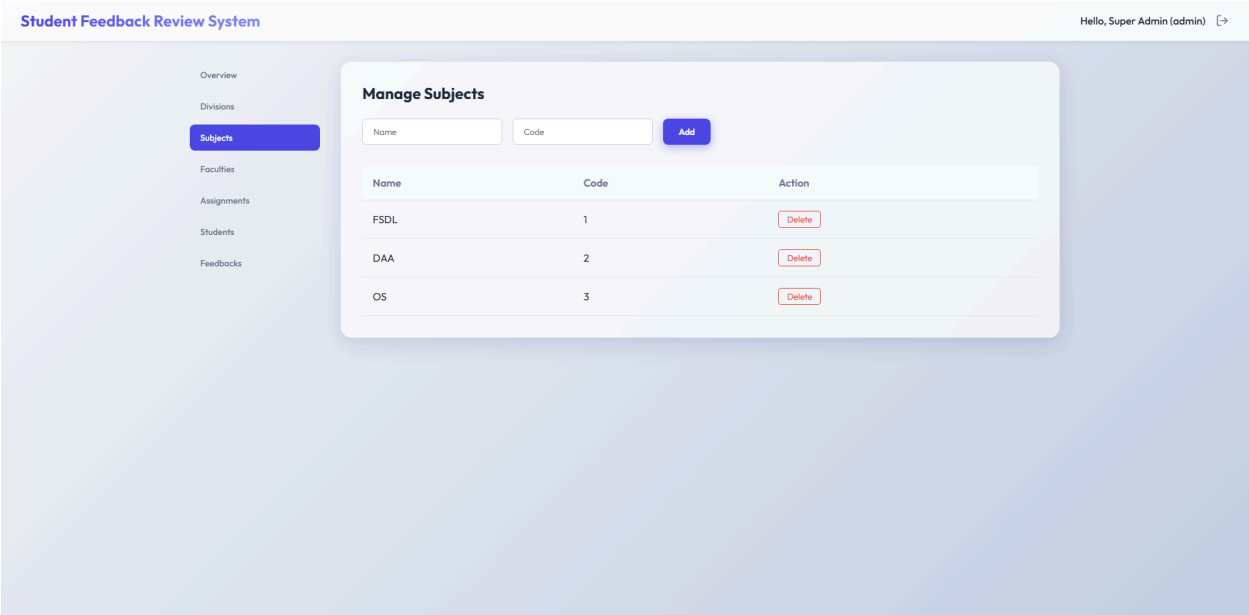
OUTPUT

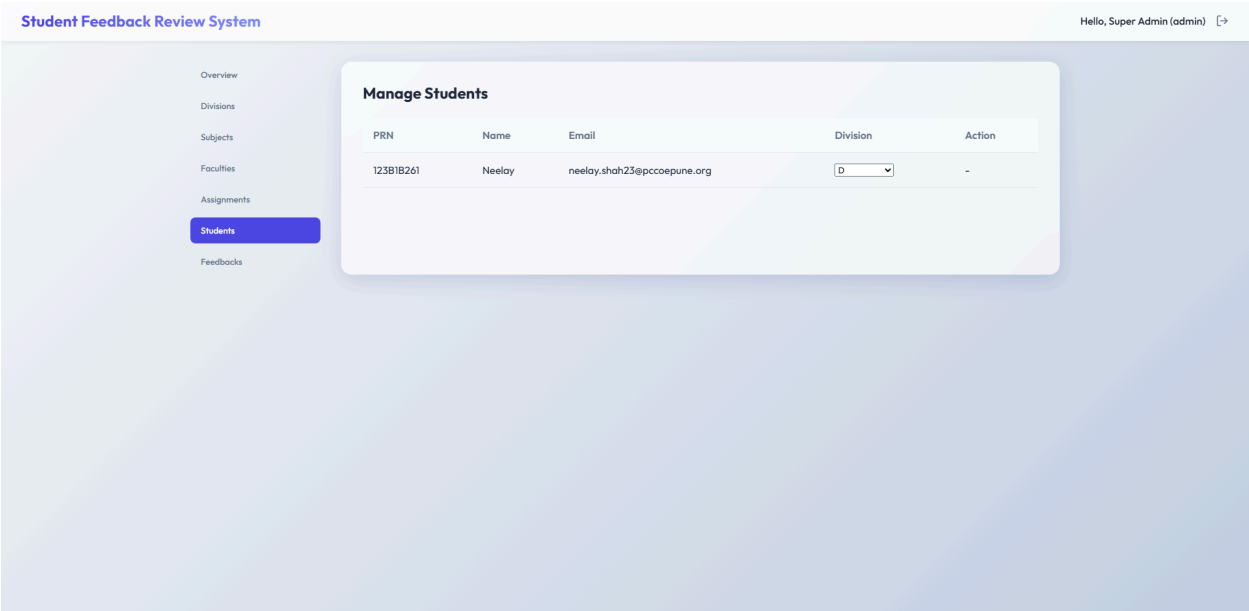



Student Feedback Review System
Student Registration
PRN (Unique ID)
123B1B261
Full Name
Neelay Shah
Email Address
neelay.shah23@pccoepune.org
Password
Register
Already have an account? Login here


Student Feedback Review System
Hello, Super Admin (admin)
Overview
Divisions
Subjects
Faculties
Assignments
Students
Feedbacks
Dashboard Overview
0
Total Feedbacks
1
Registered Students
2
Faculties
No feedback data to chart


Student Feedback Review System
Hello, Super Admin (admin)
Overview
Divisions
Subjects
Faculties
Assignments
Students
Feedbacks
Manage Divisions
Division Name
Add
Division Name
Action
A
Delete
B
Delete
C
Delete
D
Delete


Student Feedback Review System
Hello, Super Admin (admin)
Overview
Divisions
Subjects
Faculties
Assignments
Students
Feedbacks
Manage Subjects
Name
Code
Add
Name
Code
Action
FSDL
1
Delete
DAA
2
Delete
OS
3
Delete


Student Feedback Review System
Hello, Super Admin (admin)
Overview
Divisions
Subjects
Faculties
Assignments
Students
Feedbacks
Manage Students
PRN
Name
Email
Division
Action
123B1B261
Neelay
neelay.shah23@pccoepune.org
D
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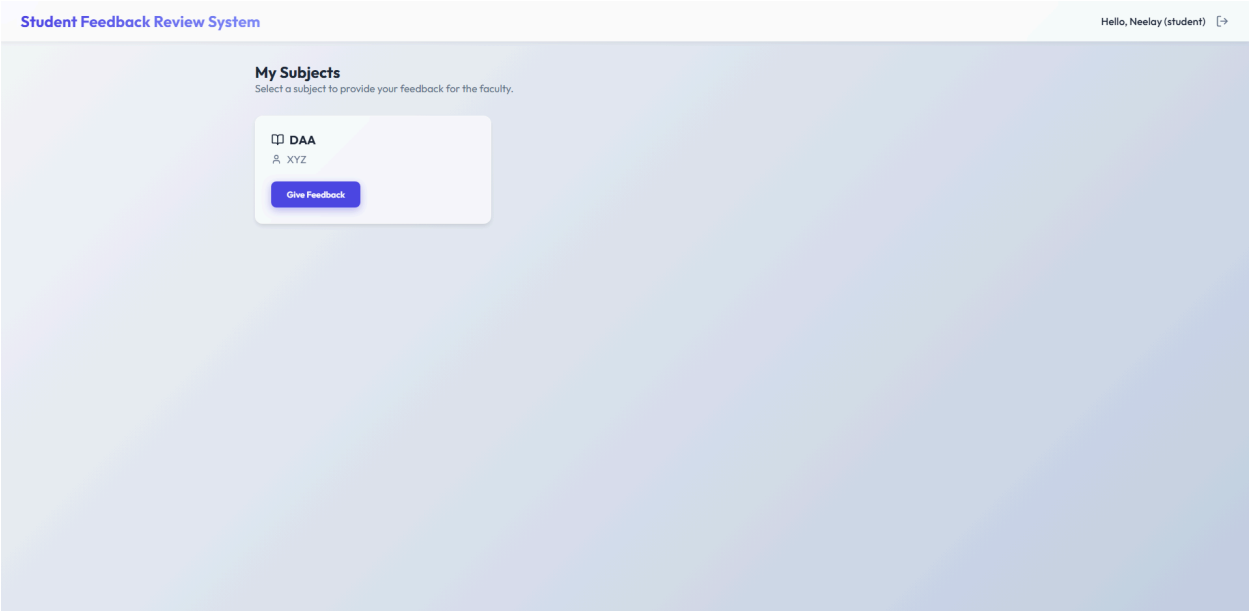
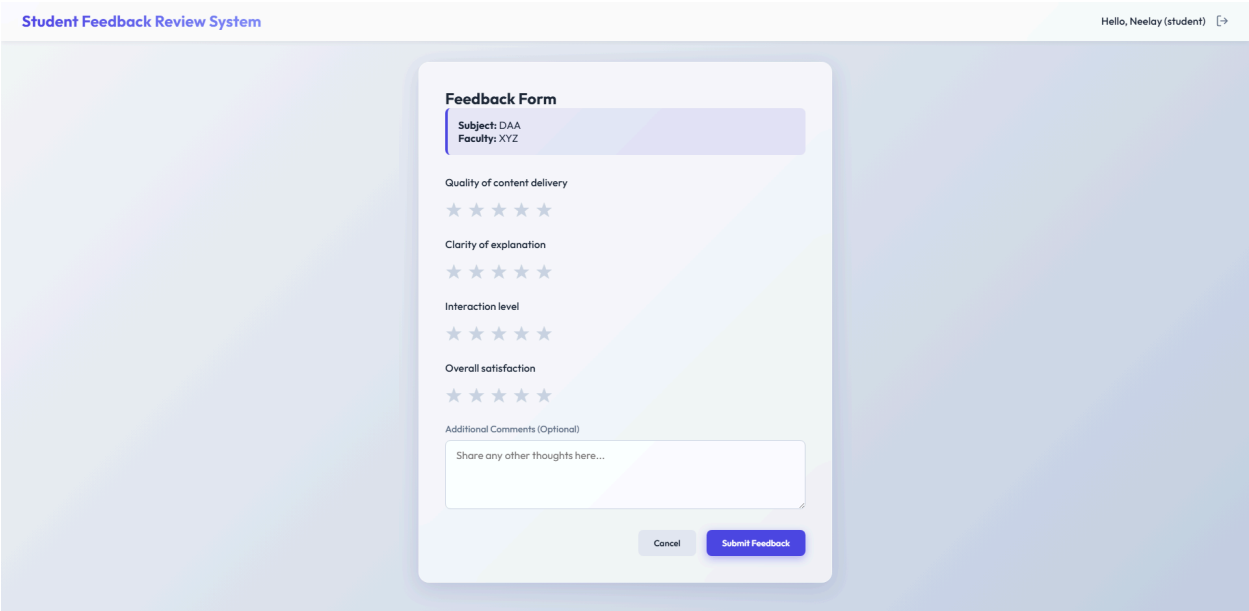
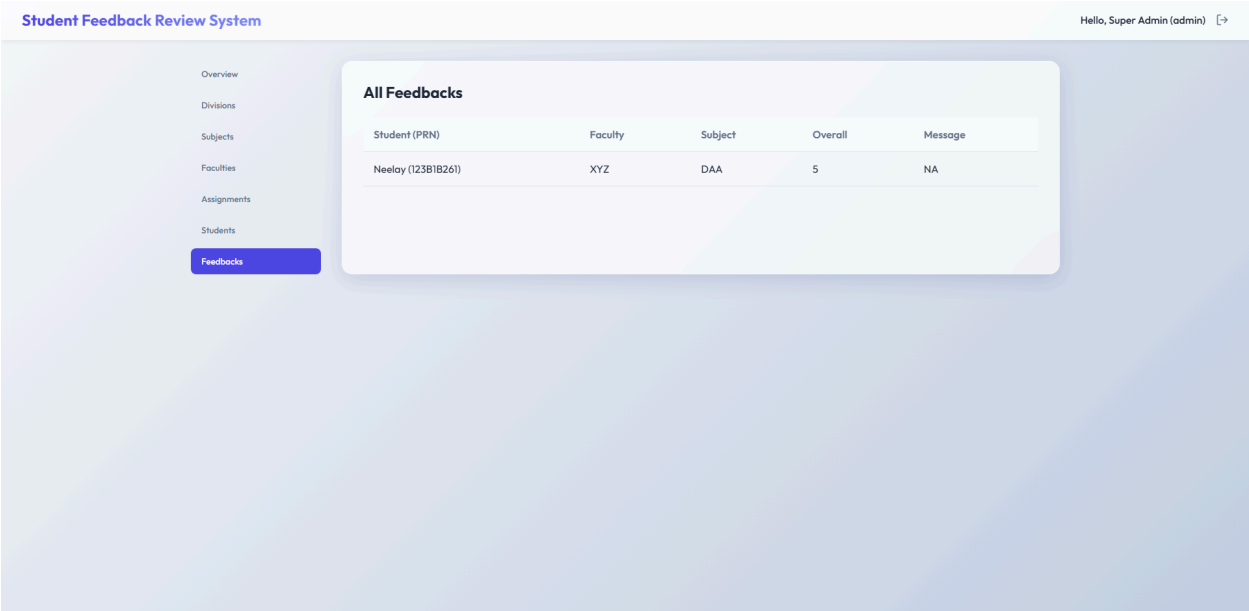

Student Feedback Review System
Hello, Neelay (student)
My Subjects
Select a subject to provide your feedback for the faculty.
DAA
XYZ
Give Feedback





Conclusion

This website offers a streamlined and efficient platform for collecting and managing course feedback using modern web technologies. It ensures secure and real-time interaction between students and the system, enabling reliable data handling. Overall, the platform supports continuous improvement in course quality and enhances the overall learning experience through structured and meaningful feedback.